This assignment will not be graded and is only for practice.

Key points: 1

The span of a set S (of vectors) is defined as the set of all finite linear combinations of elements (vectors) of S and denoted by $Span(S)$ i.e., $Span(S) = \{\sum_{i=1}^{n} a_i v_i \in V \mid a_1, a_2, ..., a_n \in \mathrm{R} \text{ and } v_1, v_2, ..., v_n \in S\}$. Let V be a vector space. A set $S \subseteq V$ is spanning set for V if $Span(S) = V$.

Algorithm to obtain a spanning set for a non-zero vector space V:

i) Start with the empty set $S_0 = \phi$.

ii) Add any non-zero vector $v_1 \in V$ to S_0 and define it as S_1. So $S_1 = S_0 \cup \{v_1\}$.

iii) If $Span(S_1) = V$, then we are done. Otherwise choose $v_2 \in V \setminus span(S_1)$ and add it to the set and call it S_2. So $S_2 = S_1 \cup \{v_2\}$.

iv) If $Span(S_2) = V$, then we are done. Otherwise, choose $v_3 \in V \setminus Span(S_2)$, and construct $S_3 = S_2 \cup \{v_3\}$.

v) Repeat the process until $Span(S_n) = V$ for some n.

1) Let $S = \{(1, 2), (2, 3), (1, -1)\} \subseteq \mathrm{R}^2$. Which of the following is (are) elements of $Span(S)$? ***1 point***

- [ ] 5(1,2) + 8(1,-1)
- [ ] (0,0)
- [ ] $2(1, 2) - 3(2, 3) + 5(1, -1)$
- [ ] $(2, 1) - 2(3, 2) - 4(-1, 1)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

5(1,2) + 8(1,-1)
(0,0)
$2(1, 2) - 3(2, 3) + 5(1, -1)$
$(2, 1) - 2(3, 2) - 4(-1, 1)$

2) Let $S = \{(1, 1, -1), (-1, 1, 1)\} \subset \mathrm{R}^3$. Which of the following is (are) elements of $Span(S)$? ***1 point***

- [ ] 2(1, 1, -1)+ (-1, 1, 1)
- [ ] (8, 8, -8)
- [ ] (0, 2,0)

☐ (1,1,-2)

No, the answer is incorrect.

Score: 0

Accepted Answers:

2(1, 1, -1)+ (-1, 1, 1)
(8, 8, -8)
(0, 2,0)

3) Let S = {(-1, 1, 1), (0, $\frac{1}{2}$,0), (0, 1, -2)}$\subset \mathrm{R}^3$. Which of the following option(s) is (are) true? ***1 point***

[**Hint:** Form the matrix using the vectors as the columns and calculate the determinant of the matrix.]

☐ S is a linearly dependent set.

☐ S is a linearly independent set.

☐ S is a basis of the vector space R^3.

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is a linearly independent set.
S is a basis of the vector space R^3.

4) Consider a set of vectors $S = \{(1,2),(-1,2),(3,1),(-3,1)\}$ in R^2. Choose the set of correct options. ***1 point***

☐ The set $\{(1,2),(-1,2),(3,1)\}$ is a basis of R^2.

☐ The set $\{(-1,2),(3,1)\}$ is a basis of R^2.

☐ The set $\{(3,1),(-3,1)\}$ is a basis of R^2.

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(-1, 2), (3, 1)\}$ is a basis of R^2.
The set $\{(3, 1), (-3, 1)\}$ is a basis of R^2.

Key points: 2

The following conditions are equivalent for a set to be a basis of a vector space:
- Maximal linearly independent set : A linearly independent set such that if any vector is appended to it, the new set is no longer linearly independent.
- Minimal spanning set : A spanning set such that if any vector is removed from the set, then the set will not be a spanning set.

5) Which of the following sets are maximal linearly independent sets in R^2? ***1 point***

- ☐ $\{(1, 0)\}$
- ☐ $\{(1, 2), (2, 4)\}$
- ☐ $\{(1, 2), (1, 1)\}$
- ☐ $\{(1, 2), (1, 1), (2, 1)\}$
- ☐ $\{(1, 0), (0, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 2), (1, 1)\}$
$\{(1, 0), (0, 1)\}$

6) Which of the following sets are minimal spanning sets of R^3? ***1 point***

- ☐ $\{(1, 0, 0), (0, 1, 0)\}$
- ☐ $\{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$
- ☐ $\{(1, 0, 1), (0, 1, 1), (1, 1, 0)\}$

- ☐ $\{(1,0,1),(0,1,1),(1,1,2)\}$
- ☐ $\{(1,0,0),(0,1,0),(0,0,1),(1,1,1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1,0,0),(1,1,0),(1,1,1)\}$
$\{(1,0,1),(0,1,1),(1,1,0)\}$

Key Points: 3

Basis: A basis of a vector space V is a linearly independent subset of V that spans V .
Basis of a vector space:

- Maximal linearly independent set.
- Minimal spanning set.

How to find a basis of a subspace of R^n?

- Suppose a subspace V of R^n is defined by some constraints, as follows:

$$V = \{(x_1, x_2, \ldots, x_n) \mid f_1(x_1, x_2, \ldots, x_n) = 0, f_2(x_1, x_2, \ldots, x_n) = 0, \ldots, f_m(x_1, x_2, \ldots, x_n) = 0\} \subseteq \mathrm{R}^n$$

Where f_i's are linear equations in general.

- **Step 1**: By solving f_i's we have to try to find out the dependent and independent variables. Suppose there are k independent variables
and there are $k - n$ dependent variables.

- **Step 2**: Choose the first independent variable, let it be x_1 and assign the value 1 to that, and assign 0 to all the other independent variables.

- **Step 3**: Find out the values of all the dependent variables with respect to the assignment given in Step 2 and call that vector to be v_1.
- **Step 4**: Carry out this process for each independent variable, to get the vectors $v_2, \ldots, v_k$.

The set of vectors $\{v_1, v_2, \ldots, v_k\}$ is a linearly independent set and this is a maximal linearly independent set in V, hence forms a basis of V. Hence, dimension of V is k in this case.

7) ***1 point***

Let S ={(-1, 1, 1), (0, $\frac{1}{2}$,0), (0, 1, -2)}$\subset \mathrm{R}^3$. Which of the following option(s) is (are) true?

Hint: Form the matrix using the vectors as the columns and calculate the determinant of the matrix.

- [ ] S is a linearly dependent set.
- [ ] S is a linearly independent set.
- [ ] S is a basis of the vector space R^3.
- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is a linearly independent set.
S is a basis of the vector space R^3.

8) Consider a set of vectors $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\}$ in R^2. Choose the set of correct options. ***1 point***

- [ ] The set { (1, 2) ,(-1, 2), (3, 1)} is a basis of R^2.
- [ ] The set { (-1, 2), (3, 1) } is a basis of R^2.
- [ ] The set { (3, 1), (-3, 1) } is a basis of R^2.
- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set { (-1, 2), (3, 1) } is a basis of R^2.
The set { (3, 1), (-3, 1) } is a basis of R^2.

9) Let V be a vector space which is defined as follows: $V = \{(x, y, z) \mid 2x + 3y = 0 \text{ and } y + 5z = 0\} \subseteq \mathrm{R}^3$ with usual addition and scalar multiplication. Which of the following sets form bases of V? (More than one option may be correct) **1 point**

[Hint: Two constraints are given here, $f_1(x, y, z) = 2x + 3y = 0$ and $f_2(x, y, z) = y + 5z = 0$. Observe that z is the independent variable and x and y are the dependent variables.]

- ☐ $\{(2, 3, 0), (0, 1, 5)\}$
- ☐ $\{(3, -2, 0), (0, 5, -1)\}$
- ☐ $\{(\frac{15}{2}, -5, 1)\}$
- ☐ $\{(15, -10, 2)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(\frac{15}{2}, -5, 1)\}$
$\{(15, -10, 2)\}$

10) Let V be a vector space which is defined as follows: $V = \{(x, y, z, w) \mid x + z = y + w\} \subseteq \mathrm{R}^4$ with usual addition and scalar multiplication. Which of the following set forms a basis of V? **1 point**

[Hint: The constraint given here is $x + z = y + w$, which can be written as $x - y + z - w = 0$. As there are 4 variables and 1 equation, the number of independent variables is 3, where as the number of dependent variable is 1. If you check the first vector of each option, only Option 2 and Option 4 satisfying the constraint, and in that vector $x = y = 1$. Start with taking y as the dependent variable, and apply the method mentioned above.]

- ○ $\{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 0, 1)\}$.
- ○ $\{(1, 1, 0, 0), (0, 1, -1, 0), (0, -1, 0, 1)\}$.
- ○ $\{(1, -1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$.
- ○ $\{(1, 1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1,1,0,0),(0,1,1,0),(0,-1,0,1)\}$.

11) Find the dimension of the vector space $V = \{(x,y,z,w) \mid x+y=z+w, x+w=y+z, \text{ and } x,y,z,w \in \mathrm{R}\}$. [Hint: Two constraints are given here $f_1(x,y,z,w) = x+y-z-w=0$ and $f_2(x,y,z,w) = x-y-z+w=0$. Try to find out the number of independent variables.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Key Points: 4

How to find a basis of a subspace V of $M_{m\times n}(\mathrm{R})$:

- **Step 1**: Consider each element of a $m \times n$ matrix as a variable. So there are mn number of variables.

- **Step 2**: Using the constraints given as the definition of the subspace separate out the independent and the dependent variables. Suppose there are k independent variables and $mn-k$ dependent variables.

- **Step 3**: Define a matrix M_1 by assigning 1 to one independent variable and 0 to all the other independent variables, together with finding out the values of all the dependent variables.

- **Step 4**: Repeat this process for all the independent variables to get the vectors $M_2, M_3, \ldots, M_k$.

The set of matrices $\{M_1, M_2, \ldots, M_k\}$ forms a basis of V. Hence the dimension of V is k in this case.

12) Let $V = \{A \mid A \in M_{3\times3}(\mathrm{R}), A \text{ is a diagonal matrix } \}$. Which of the following sets form a ***1 point*** basis of V?

[Hint: $a_{ij} = 0$ if $i = j$. Thus there are 6 constraints and these are the dependent variables while the diagonal entries are the 3 independent elements.]

- $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$
- $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$
- $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$
- $\left\{ \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \right\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$

13) Let $V = \{A \mid A \in M_{3\times3}(\mathrm{R}), A \text{ is a diagonal matrix and sum of the diagonal elements are } 0 \}$. What will be the dimension of V?

[Hint: The general form a diagonal matrix of order 3 is $\begin{bmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{bmatrix}$. Hence there are three variables, a_1, a_2, a_3. Now there is one more constraint, which says that the sum of the diagonal elements is 0. Hence we have $a_1 + a_2 + a_3 = 0$. So we can choose a_3 to be the dependent variable and a_1 and a_2 to be the independent variables, as $a_3 = -a_1 - a_2$.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Key points: 5

Rank of a matrix:

- Maximum number of linearly independent column vectors (or row vectors) in a matrix.

NOTE: Rank of a matrix is always less than or equal to the number of columns (or rows).

How to find rank of a matrix A:

- Reduce the matrix to row echelon form.
- Find the number of non zero rows in the reduced matrix which will be the rank of the matrix A.

14) Consider the following two matrices: ***1 point***

$$P = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -2 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 1 \end{bmatrix}; Q = \begin{bmatrix} 1 & 1 & -2 & 0 \\ 1 & 0 & 2 & 0 \\ 0 & -1 & 0 & 4 \end{bmatrix}$$

Which of the following statements is (are) TRUE? [Hint: Try to reduce the matrices to row echelon form and find the number of non zero rows]

- ☐ Rank(P) = 4
- ☐ Rank(P) = Rank(Q)
- ☐ Rank(QP) = 4
- ☐ Rank(PQ) = Rank(QP)

No, the answer is incorrect.

Score: 0

Accepted Answers:

Rank(P) = Rank(Q)

Rank(PQ) = Rank(QP)

Your score is: 0/14